

2.5 Exercises

- 2.1 There is no difference between information and data. True or False?
- 2.2 For categorical data such as the type of fruits, the one-hot encoding is preferred over an arbitrary number assignment such as one for orange, two for apple, three for mango, etc. True or False?
- 2.3 Which of the following represents the characteristics of ordinal data?
 - (a) Ordinal data has a median
 - (b) Value of interval is unknown
 - (c) Measures non-numeric traits such as satisfaction, happiness, etc.
 - (d) Establish a relative rank
 - (e) None of these
- 2.4 Determine whether the following items belong to interval or ratio data?
 - (a) The number of hours one spent studying each day
 - (b) The score of an intelligence test
 - (c) The temperatures in cities throughout the country that are listed in most newspapers
 - (d) The birth weights of babies who were born at Samsung Hospital last week
 - (e) Ages of students in a machine learning class
 - (f) Number of pages in your statistic book
 - (g) The annual salaries for all teachers in a town
 - (h) The cost of a pair of shoes
- 2.5 Which of the following is incorrect regarding the properties of distance metrics?
 - (a) The distance from $\mathbf{x}$ to $\mathbf{y}$ is the same as the distance from $\mathbf{y}$ to $\mathbf{x}$
 - (b) The metric must satisfy the triangular inequality
 - (c) The distance between two identical vectors, $\mathbf{x}$ and $\mathbf{y}$, is a non-zero value
 - (d) The distance must never be negative
 - (e) None of these
- 2.6 Which of the following distance measures calculates the distance between two binary vectors?
 - (a) Euclidean distance
 - (b) Manhattan distance
 - (c) Minkowski distance
 - (d) Hamming distance
 - (e) None of these
- 2.7 What is the preferred way to handle missing or corrupted data in a small data set?
 - (a) Drop the entire data sample that contains the corrupted features
 - (b) Replace missing values with mean/median/mode
 - (c) Replace missing features with zeros

- (d) Do not change the corrupted features
- (e) None of these

- 2.8 When the bounds of data are not known, the z-score standardization is preferred over min-max normalization. True or False?
- 2.9 You are given a set of data for supervised learning. A sample block of data looks like this:

```

1.0234, 0.3202, 25, 0.0081, 60033.81, 3
1.1356, 0.3308, 21, 0.0067, 69283.18, -1
0.9988, 0.2526, 23, 0.0093, 66034.33, 2
1.1858, 0.3001, 22, 0.0077, 64037.35, 1
1.1533, 0.3853, 21, 0.0066, 62033.58, -13
1.0755, 0.3102, 21, 0.0098, 60183.65, 1
1.0045, 0.2901, 22, 0.0065, 61093.98, -2
1.1031, 0.3912, 21, 0.0088, 69033.23, -1

```

Each row corresponds to a sample data measurement with 5 input features and 1 response. What kind of undesired effect can you anticipate if this set of raw data is used for learning?

- (a) Those features with nominal values may create numerical instability
 - (b) Those features with very large values may overshadow those with very small values
 - (c) Those features with integer values may add bias to the system
 - (d) The sample with output response value -13 should be imputed
 - (e) None of these
- 2.10 Suppose you are given the following data regarding the monthly income of participants of a yoga learning group.

Participants	a	b	c	d	e	f	g	h	i
Income \$	5000	3000	4000	4500	6000	3500	5000	800000	5500

Among mean, mode, and median, which is the most appropriate statistic to describe the data? Provide the value of this statistic based on the given table.